

Evaluating AI's Role in Public Technology Literacy in Ghana

A Mixed-Methods Research Protocol and Conceptual Framework

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Abstract

Artificial intelligence (AI) is becoming embedded in everyday digital services, education, communication, and commerce in Ghana, but widespread use does not necessarily imply meaningful technology literacy. This research protocol proposes a mixed-methods evaluation of whether AI-supported digital-literacy training can improve citizens' practical digital skills, critical evaluation of information, cybersecurity and privacy awareness, responsible AI use, and digital independence. The study conceptualizes public AI literacy across five dimensions: access, functional skills, critical AI literacy, safety and ethical use, and empowerment/outcomes. A quasi-experimental pre-test/post-test design is proposed, comparing an AI-supported training group with a conventional digital-literacy comparison group across diverse urban, peri-urban, rural, and connectivity-constrained settings in Ghana. Proposed data sources include surveys, practical skills assessments, focus groups, stakeholder interviews, and direct observation of training sessions. A six-week community-based intervention is outlined. The current target sample of approximately 300-600 participants is provisional and will be replaced by a formal sample-size/power justification before recruitment. The protocol is motivated by Ghana's rapid digital adoption, persistent inclusion gaps, growing cybersecurity and misinformation risks, and the launch of the country's National AI Strategy in April 2026. The intended contribution is a locally grounded framework for assessing whether AI makes citizens safer, more critical, more capable, and more independent technology users rather than simply more frequent users of digital tools.

Keywords: artificial intelligence; AI literacy; digital literacy; Ghana; digital inclusion; cybersecurity awareness; misinformation; responsible AI; technology education

1. Introduction

Digital technology is now deeply embedded in everyday life in Ghana through mobile money, messaging platforms, online learning, public services, and increasingly AI-enabled tools. Ghana's internet uptake increased from about 8% in 2010 to 69% in 2021, while urban uptake reached about 80% compared with roughly 54% in rural areas (Dabalen & Mensah, 2023). Smartphone access dominates internet use, underscoring the importance of mobile-first approaches to digital learning.

Yet technology use and technology literacy are not the same. A person may be able to open an application, use a chatbot, or complete a digital transaction without understanding whether an AI-generated answer is accurate, how personal data may be processed, how algorithmic systems shape information, how synthetic media can be used for fraud, or when a human review is necessary. This study therefore focuses on whether AI-supported learning can improve not only access and convenience but also judgement, safety, verification habits, and digital independence.

The need is especially relevant because digital participation remains uneven. Research using Ghana's 2021 Population and Housing Census found that 52.9% of older adults had used the internet in the preceding three months, mainly through mobile phones, and that use varied substantially by age, education, residence, disability, and household characteristics (Kyei-Arthur, 2024). These differences imply that any national AI-literacy approach must account for accessibility, affordability, language, and unequal digital confidence.

2. Policy and cybersecurity context

Ghana launched its first National Artificial Intelligence Strategy on 24 April 2026. Government and UNESCO descriptions frame the strategy as a decade-long national blueprint spanning areas such as education, infrastructure, data governance, research, public-sector transformation, and responsible AI (Ministry of Communication, Digital Technology and Innovations, 2026; UNESCO, 2026). This creates a policy window for measuring whether national AI ambitions are translating into public capability.

Cybersecurity conditions make that measurement urgent. Ghana's Cyber Security Authority has warned about AI-generated deepfake videos used to promote fraudulent investments and products and has emphasized misinformation, media literacy, fact-checking, and responsible technology use in national awareness programmes (Cyber Security Authority, 2025a, 2025b). In this environment, public AI literacy should include not only the ability to use AI tools but also the ability to verify, challenge, and safely limit their use.

3. Research problem

AI adoption may be outpacing public understanding. The central research problem is whether AI helps Ghanaians become more capable, critical, safe, and independent technology users, or whether it mainly makes digital services easier to consume without improving understanding.

The project therefore distinguishes between digital use and digital literacy. The latter requires the ability to judge output quality, verify claims, protect personal information, recognize manipulation and fraud, understand important limitations, and know when to seek human assistance or review.

4. Aim and objectives

Aim: To evaluate the role of artificial intelligence in improving public technology literacy in Ghana, with attention to access, skills, critical thinking, safety, inclusion, and responsible use.

1. Assess public awareness and use of AI-enabled tools in Ghana.
2. Measure whether AI-assisted learning improves practical digital skills.
3. Examine how AI affects citizens' ability to assess information critically.
4. Identify barriers that keep disadvantaged groups from benefiting from AI-literacy programmes.
5. Evaluate public understanding of AI-related privacy, cybersecurity, bias, and misinformation risks.
6. Compare AI-supported training against conventional digital-literacy training.

7. Develop evidence-informed recommendations for government, schools, civil society, libraries, community centres, and technology providers.

5. Research questions

8. What level of awareness do Ghanaian citizens have of AI and AI-enabled services?
9. Which groups are most likely to use AI tools, and which are excluded?
10. Does AI-supported learning improve citizens' ability to perform practical digital tasks?
11. Does AI-supported learning improve media literacy and fact-checking behaviour?
12. Can citizens recognize the risks of deepfakes, phishing, automated scams, privacy breaches, and biased AI outputs?
13. How do language, location, income, gender, age, disability, education, device ownership, and data cost affect AI literacy?
14. What types of AI learning tools work best for Ghanaian communities?
15. How can Ghana build an inclusive, ethical, and locally relevant model of AI literacy?

6. Conceptual framework: five dimensions of public AI literacy

Access. Whether people can reach and use AI-enabled tools, including device and internet availability, data and electricity costs, local-language access, disability inclusion, and rural-urban participation.

Functional skills. Whether participants can complete practical digital tasks such as managing accounts, locating reliable information, navigating e-government services, making digital transactions safely, and using AI for assistance while checking its output.

Critical AI literacy. Whether people understand that AI output is not automatically true, neutral, private, or fair, and whether they can question sources, data, limitations, possible bias, and the need for human judgement.

Safety and ethical use. Whether people can protect themselves and others through secure practices, phishing recognition, safe data handling, synthetic-media awareness, consent, and responsible use in school, work, and business.

Empowerment and outcomes. Whether increased literacy translates into practical independence, safer behaviour, improved access to services and opportunities, reduced vulnerability to fraud, and stronger participation in digital civic and economic life.

7. Proposed methodology

7.1 Study design

A mixed-methods, quasi-experimental pre-test/post-test design is proposed. The intervention group would receive AI-supported digital-literacy training, while the comparison group would receive conventional digital-literacy training without AI tools or receive the AI programme later. The design is intended to separate the effect of AI-supported learning from the effect of training in general.

7.2 Study settings and participants

Study sites should reflect Ghana's diversity, with representation from an urban setting (for example, Greater Accra or Kumasi), a peri-urban municipality, a rural community in the northern, middle, or coastal belt, and a

community with limited connectivity. Potential participant groups include students, teachers, small-business owners and informal-sector workers, public servants, market traders, parents and caregivers, older adults, persons with disabilities, and rural or local-language-speaking residents.

The draft project proposes approximately 300-600 participants across sites. This range is provisional. Before recruitment, the final sample size should be justified using a formal power calculation or another transparent design-based rationale tied to the primary outcomes, expected effect size, group allocation, and planned subgroup analyses.

7.3 Data collection

- Survey: demographics; device and internet access; AI-use frequency; basic digital skills; understanding of AI concepts; perceived benefits and risks; privacy practices; and confidence using online services.
- Practical skills assessment: tasks such as locating an official public service, judging whether a website is legitimate, asking an AI tool for an explanation and identifying possible errors, identifying phishing, and verifying an AI-generated claim against a credible independent source.
- Focus groups: trust and distrust, employment concerns, local-language barriers, school-use concerns, prior scam experiences, access constraints, and gendered or disability-related barriers.
- Stakeholder interviews: teachers, ICT trainers, cybersecurity professionals, local-government officials, civil-society representatives, community leaders, and, where feasible, relevant public agencies.
- Direct observation: connectivity, language of instruction, device-sharing, participation patterns, and whether trainers explain AI limitations as well as features.

7.4 Six-week intervention

Week	Topic	Practical outcome
1	Digital foundations	Device basics, internet navigation, and account security
2	Understanding AI	What AI is, where it is used, and what it cannot reliably do
3	AI for learning and work	Using AI for CV writing, basic research, translation, and business communication
4	Information verification	Comparing AI answers with credible sources and spotting misinformation
5	Privacy and cybersecurity	Identifying phishing, unsafe links, fraud, and risky data sharing
6	Responsible AI use	Discussing bias, fairness, consent, intellectual property, and community impact

Delivery should be practical, mobile-first, low-data where possible, locally contextualized, and supported by trusted community institutions. Local-language explanation and peer support should be incorporated where feasible.

7.5 Outcome measures

Candidate primary and secondary outcomes include practical digital-skills scores, the ability to recognize that AI can produce false information, phishing-recognition performance, safe-data-handling knowledge, independent verification behaviour, and self-reported changes in safer online behaviour. Subgroup differences can be examined by age, gender, education, location, disability, income, device ownership, and language.

A working operational definition of an AI-literate participant is someone who can explain at least one useful and one risky use of AI, use an AI tool to complete a simple task, verify an AI-generated answer against an independent source, identify a privacy risk, recognize a likely AI-enabled scam, and know when to seek human advice instead of relying on the tool.

7.6 Analysis plan

Quantitative analysis is expected to include descriptive statistics, pre/post comparisons between intervention and comparison groups, cross-tabulation across key demographic and access variables, and regression modelling to identify predictors of AI-literacy outcomes. The final model specification, covariates, treatment of missing data, and multiplicity strategy should be fixed in the preregistration before analysis begins.

Qualitative interview and focus-group data would be analysed thematically, with attention to recurring themes such as trust, fear, language, privacy, scams, employment, access, and community support. The final coding approach and procedures for resolving coding disagreements should be documented before formal analysis.

7.7 Ethics, safeguarding, and data protection

No participant data have been collected for this manuscript version. Before recruitment, the study should obtain approval or formal determination from an appropriate ethics review body and finalize informed-consent procedures. If minors are included, parental/guardian consent, participant assent, and additional safeguarding procedures will be required.

The study should minimize collection of personally identifiable information, avoid collecting passwords or financial credentials, separate contact details from research responses where possible, and use access controls and de-identification for stored data. Participants should be clearly told not to enter sensitive personal information into public AI systems during study tasks.

8. Rationale for local relevance

A Ghanaian AI-literacy model should be designed for the country's mobile-first reality and should not assume stable broadband, paid AI subscriptions, English fluency, or equal device ownership. Existing initiatives indicate that AI capacity-building can be delivered through education systems: Africa ICT Right reported seven three-day AI and digital-pedagogy trainings reaching 29 District STEM Coordinators and 466 STEM teacher educators, alongside collaboration with the Ghana Education Service STEM Directorate (Team4Tech, 2026).

The research therefore treats language, infrastructure, affordability, disability inclusion, and community trust as core design variables rather than peripheral considerations.

9. Anticipated contribution and policy relevance

The study is intended to contribute a measurable, locally grounded model of public AI literacy that can inform digital-inclusion programmes, schools and universities, community training, cybersecurity awareness, and technology-provider design. The study does not assume that AI will improve literacy; rather, it is designed to

test whether AI-supported training produces better outcomes than conventional digital-literacy training and for whom.

Potential policy relevance includes the design of public AI-literacy programmes, local-language and low-data learning tools, practical cybersecurity and misinformation education, and mechanisms for human review when automated systems affect citizens.

10. Limitations and items to finalize before data collection

- The proposed 300-600 participant range is provisional and must be replaced by a formal sample-size justification.
- Exact study districts, recruitment channels, and site-selection criteria remain to be finalized.
- The participant age range has not yet been fixed; inclusion of minors would require additional ethics and safeguarding procedures.
- The exact intervention/comparison allocation procedure is not yet specified.
- Primary versus secondary outcomes, the final statistical model, missing-data rules, and exclusion criteria must be fixed before preregistration is submitted.
- No empirical results are reported in this version; all claims about likely outcomes are hypotheses or policy motivations.

11. Research status and open-science plan

This manuscript is a research protocol and conceptual framework. The planned next steps are to finalize the preregistration, obtain appropriate ethics review, pilot the survey and practical-skills instruments, finalize the sample-size justification, and then begin data collection. Study materials and version history may be shared through an open project repository, subject to privacy, ethics, and intellectual-property constraints.

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